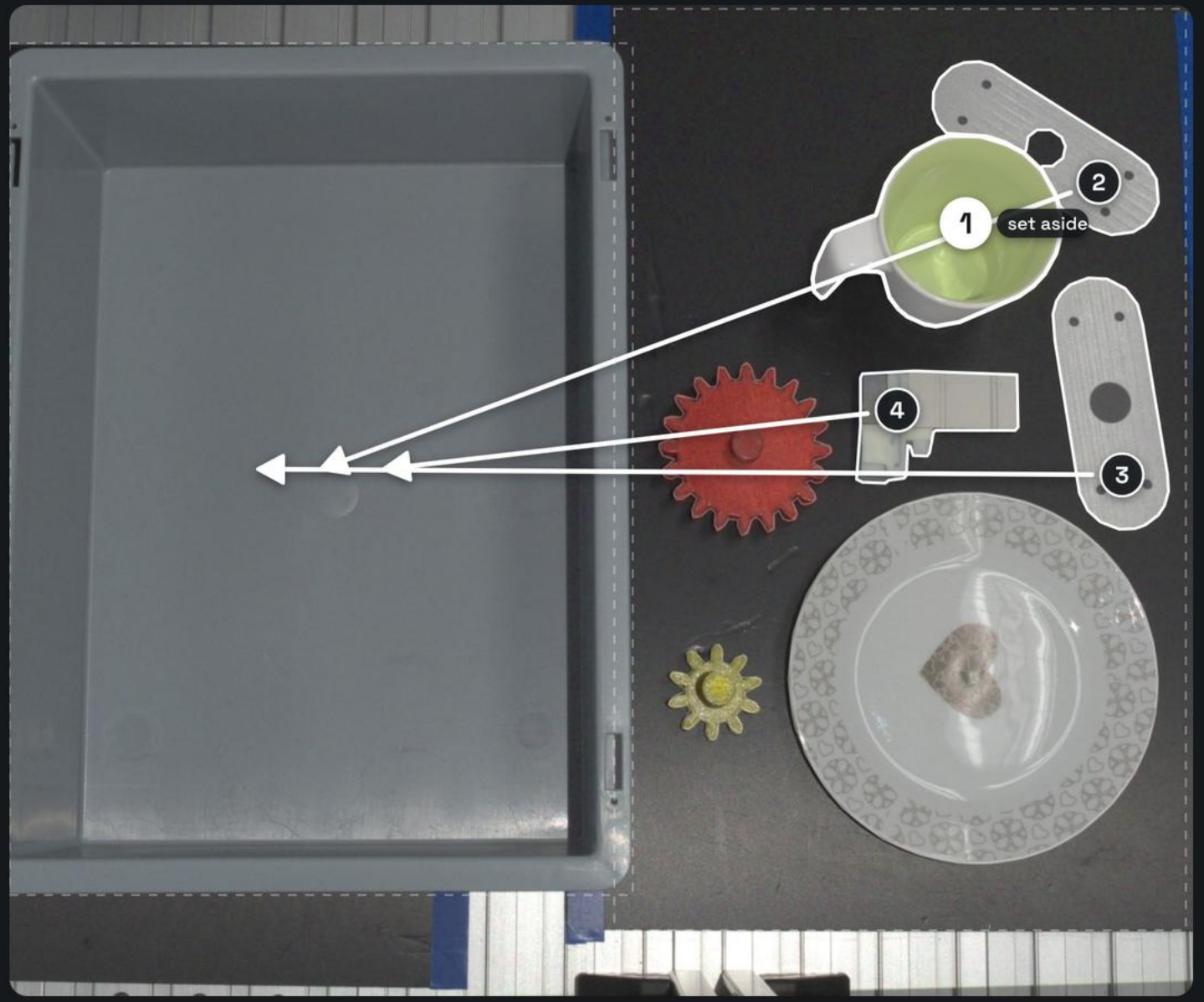
See it.
Say it.
Pick it.

Apertus plans on physics, not pixels.

We measure widths in millimetres, what lies on what, and which tray it's in. Apertus decides on those facts, and our check holds it to them.



Team Claw & Order
Daniel Alznauer, Giuseppe Prezzo

Hack Apertus 2026, ZHAW challenge

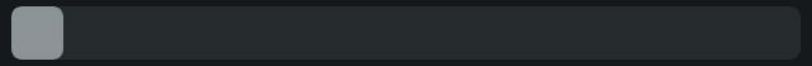
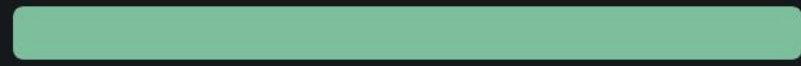
We turn the photo into physical facts

Example, public scene 7: **“put the red gear under the mug in the black tray”**

1	**See**	Grounding DINO, SAM 2.1 and SigLIP 2	Red gear 1, green mug 1, red cube 1, yellow gear 1.
2	**Measure**	Our code, using the camera calibration	red_gear_1: 67 mm wide, can be picked. Lies under green_mug_1.
3	**Decide**	Apertus 8B, reading only the facts	Pick red gear 1, put it in the black tray.
4	**Check**	Our code, against the same facts	**Error** The mug still lies on red gear 1. Move it first. **Second try** Mug to the table, then red gear 1. Accepted.

Open models only, run locally. One command for any folder of scenes, with a Docker image.

Physics, not pixels: 64.1 of 65

	Apertus alone looks at the photo and points itself	**Our pipeline** Apertus plans on measured facts
Grounding 35 points: is the first pick the right object?	 **7.9** / 35 Right object in 9 of 40 plans.	 **34.1** / 35 Right object in 39 of 40 plans.
Safety 30 points: does it refuse when it must?	 **2.0** / 30 1 of 10 right, 13 of 40 doable refused.	 **30.0** / 30 10 of 10 right, none of 40 doable refused.
Total 65 points	**9.9 / 65**	**64.1 / 65**

These 50 instructions are public and we tuned on them. On 405 new instructions written afterwards: 96 % right.

Where Apertus needed help

1

It picks what isn't there

Asked for "the yellow cube" with only a yellow gear in view, it takes the gear.

Our fix

When in doubt, don't move: if the retries run out and Apertus last refused, the refusal stands. Correct refusals 31 → 40 of 56.

2

It ignores facts handed to it

Told the measured distances in a correction, it repeats the error in 98 % of retries.

Our fix

Code computes every relation up front. Apertus only chooses.

3

Its first reply rarely holds

Only 39 % of first replies pass the checks.

Our fix

The check names the violation and asks again, at most twice.

For re-runs: Apertus overflows in float16. Use bfloat16, or float32 on older GPUs.

Measured on 405 instructions of our own and on the held-out run.

Facts and checks change Apertus's decisions. More prompt text doesn't.

We tried three additions to the prompt: the facts up front, a worked example, a refusal example. None changed Apertus's decisions. What did: code that names the exact violation.

Next

Accuracy: train, don't prompt

One LoRA pass: right first time 70 → 84 of 126, but 3 new wrong refusals. Next: learn from the check's rejections too.

Production: one small model per cell

A quantised Apertus with the cell's adapter on an 8 GB GPU. Open: does quantisation change decisions?

Code Apache 2.0, dataset CC BY 4.0. Team Claw & Order.